



Mathematics Part 1 — Cbse Half-Yearly Exam — 80 Marks

Subject: Mathematics Part 1 | Class: Class 9

Total Marks: 80 | Time Allowed: 3h | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 20 questions. Each carries 1 mark(s).		
1	According to the concepts of the number system, what does the union of rational and irrational numbers form? (A) The set of Imaginary Numbers (B) The continuous, unbroken Real Number Line (C) The set of Integers (D) The cyclic number system <i>[Ch 3: The World of Numbers]</i>	1
2	Assertion (A): The sample space for rolling a standard die is {1, 2, 3, 4, 5, 6}. Reason (R): The sample space is the set of all possible outcomes in an experiment. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 7: The Mathematics of Maybe: Introduction to Probability]</i>	1
3	A ball is dropped from a height of 24 feet. Each time it bounces, it reaches $\frac{3}{4}$ of its previous height. What is the maximum height attained after the second bounce? (A) 18 feet (B) 10.125 feet (C) 7.59375 feet (D) 13.5 feet <i>[Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]</i>	1



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4	Which of the following correctly defines an explicit formula for a sequence? (A) A rule that gives the value of a term using the values of previous terms (B) A rule that uses the term's position number n to calculate its value (C) A rule that finds the sum of all terms in the sequence (D) A rule that calculates the difference between consecutive terms <i>[Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]</i>	1
5	What is a quadrilateral whose vertices are concyclic called? (A) Symmetric quadrilateral (B) Isosceles quadrilateral (C) Equilateral quadrilateral (D) Cyclic quadrilateral <i>[Ch 5: I'm Up and Down, and Round and Round]</i>	1
6	Which of the following are lines of reflection symmetry for a circle? (A) Only the horizontal and vertical diameters (B) Only the radii (C) All diameters (D) All chords <i>[Ch 5: I'm Up and Down, and Round and Round]</i>	1
7	What is the coefficient of z in the polynomial $4z^3 + 5z^2 - 11$? (A) 5 (B) -11 (C) 0 (D) 4 <i>[Ch 2: Introduction to Linear Polynomials]</i>	1
8	Assertion (A): The numerator $x^2 - 7x + 12$ of a rational expression factors as $(x - 3)(x - 4)$. Reason (R): The denominator $5x^2 + 5x - 100$ of the same rational expression factors as $5(x - 4)(x + 5)$. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 4: Exploring Algebraic Identities]</i>	1



Q.NO.	QUESTIONS	MARKS
9	If $\sqrt{2} = p/q$ leads to $2q^2 = p^2$, and we let $p = 2k$ (since p must be even), what is the resulting equation after substitution and simplification? (A) $q^2 = k^2$ (B) $2q^2 = 2k^2$ (C) $q^2 = 2k^2$ (D) $q^2 = 4k^2$ <i>[Ch 3: The World of Numbers]</i>	1
10	Assertion (A): For an obtuse-angled triangle, the circumcentre lies inside the triangle. Reason (R): The circumcentre of a right-angled triangle lies at the midpoint of the hypotenuse. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 5: I'm Up and Down, and Round and Round]</i>	1
11	Assertion (A): Brahmagupta's formula can be applied to any quadrilateral to find its area. Reason (R): Brahmagupta's formula generalises Heron's formula. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 6: Measuring Space: Perimeter and Area]</i>	1
12	What is the unit of measurement for area, based on a 1×1 square? (A) 1 sq. unit (B) 1 unit (C) 1 cubic unit (D) 1 perimeter unit <i>[Ch 6: Measuring Space: Perimeter and Area]</i>	1
13	Assertion (A): A tree diagram is useful for visualising multi-step experiments where each path from start to end represents one complete outcome. Reason (R): A multi-step experiment involves a series of independent trials. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 7: The Mathematics of Maybe: Introduction to Probability]</i>	1



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14	The Babylonians found through measurement that for a circle, $C^2/A \approx 12$. What formula for the area of a circle does this yield? (A) $A \approx C^2/12$ (B) $A \approx 12C$ (C) $A \approx C/12$ (D) $A \approx 12C^2$ <i>[Ch 6: Measuring Space: Perimeter and Area]</i>	1
15	The straight lines representing an equation of the form $y = ax$ always pass through which point? (A) (0, 0) (B) (1, 1) (C) (a, 0) (D) (0, a) <i>[Ch 2: Introduction to Linear Polynomials]</i>	1
16	What does it mean for a coin to be considered 'fair' or 'unbiased'? (A) It is symmetrical so there is no reason for it to come down more often on one side. (B) It has heads on both sides. (C) It always lands on tails. (D) It is made of a specific metal. <i>[Ch 7: The Mathematics of Maybe: Introduction to Probability]</i>	1
17	What is the difference between an algebraic identity and an equation? (A) An identity is true for all values of the variables, while an equation need not be true for all values. (B) An equation is true for all values of the variables, while an identity is true for only one value. (C) Identities can only have one variable, while equations can have multiple variables. (D) There is no difference between an identity and an equation. <i>[Ch 4: Exploring Algebraic Identities]</i>	1
18	Who formalised the notion and use of zero and negative numbers as algebraic entities, making the four-quadrant Cartesian plane possible? (A) Brahmagupta (B) René Descartes (C) Al-Bīrūnī (D) Ōmar Khayyām <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	1



Q.NO.	QUESTIONS	MARKS
19	What is the x-coordinate of a point that lies on the y-axis? (A) 0 (B) 1 (C) -1 (D) Undefined <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	1
20	The letters of the word 'PEACE' are placed on cards, and one card is drawn without looking. What is the probability that the card is not an E? (A) $\frac{3}{5}$ (B) $\frac{2}{5}$ (C) $\frac{1}{5}$ (D) $\frac{4}{5}$ <i>[Ch 7: The Mathematics of Maybe: Introduction to Probability]</i>	1
Section B Attempt 5 out of 7.		
21	Describe the construction steps to locate $\sqrt{2}$ on the number line using a unit square's diagonal. <i>[Ch 3: The World of Numbers]</i>	2
	OR Find the area of a quadrant of a circle whose circumference is 44 cm. (Use $\pi = \frac{22}{7}$) <i>[Ch 6: Measuring Space: Perimeter and Area]</i>	
22	In a survey of 50 students, 15 students said they liked football. The number of students who like football is 15, and the ____ (frequency/relative frequency) is ____ (fill in the fraction or decimal). <i>[Ch 7: The Mathematics of Maybe: Introduction to Probability]</i>	2
	OR An arc of a circle subtends an angle of 110° at the centre. What is the measure of the angle subtended by this arc at a point on the circle outside the arc? <i>[Ch 5: I'm Up and Down, and Round and Round]</i>	
23	Bela has ₹100 for pocket money and spends ₹5 every day. Write a linear expression for the amount of money left on the n^{th} day. <i>[Ch 2: Introduction to Linear Polynomials]</i>	2
24	What is the side of the cube whose volume is $p^3 + 6p^2q + 12pq^2 + 8q^3$ cubic units? <i>[Ch 4: Exploring Algebraic Identities]</i>	2
25	State the general formula for the distance between two points (x_1, y_1) and (x_2, y_2) in the 2-D plane.	2



Q.NO.	QUESTIONS	MARKS
	<i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	
Section C Attempt 5 out of 7.		
26	Describe the steps to construct a line segment of length $\sqrt{2}$ on the number line using a ruler and compass. <i>[Ch 3: The World of Numbers]</i>	3
	OR When $d = 0$ in a 4-gon with sides a, b, c, d , show that the semi-perimeter of the 4-gon equals the semi-perimeter of the triangle with sides a, b, c . <i>[Ch 6: Measuring Space: Perimeter and Area]</i>	
27	Find the value of the quadratic polynomial $7s^2 - 4s + 6$ when $s = -3$. <i>[Ch 2: Introduction to Linear Polynomials]</i>	3
	OR When triangle ADM is reflected in the y -axis, the image of point $A(3, 4)$ is A' and the image of point $D(7, 1)$ is D' . Calculate the distance $A'D'$ using the coordinates of the reflected points. <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	
28	Define an event in the context of probability. If a standard 6-sided die is rolled once, write the event for 'rolling a number greater than 4' and list its elements. <i>[Ch 7: The Mathematics of Maybe: Introduction to Probability]</i>	3
29	An arc AFB of a circle with centre C subtends an angle of 120° at the centre. Find the angle subtended by this arc at a point D on the circle outside the arc. <i>[Ch 5: I'm Up and Down, and Round and Round]</i>	3
30	Prove that for any three consecutive square numbers $(n-1)^2, n^2$, and $(n+1)^2$, the sum of the smallest and largest squares minus two times the middle square always equals 2. <i>[Ch 4: Exploring Algebraic Identities]</i>	3
Section D Attempt 4 out of 6.		
31	Brahmagupta introduced the concepts of "fortunes" (Dhana) and "debts" (Riña) to represent positive and negative numbers. Using these real-world concepts, explain why: (i) the product of two debts is a fortune, and (ii) subtracting a negative number is the same as adding a positive number. Provide a numerical example for each. <i>[Ch 3: The World of Numbers]</i>	5



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	OR Explain how the area of a circle formula $A = \pi r^2$ can be derived by slicing the circle and rearranging the slices into a parallelogram-like shape. Also, state how Archimedes expressed the area of a circle. <i>[Ch 6: Measuring Space: Perimeter and Area]</i>	
32	State the identity for $(a + b + c)^2$ and describe its geometric interpretation using a square of side $(a + b + c)$. Use this identity to find the value of 119^2 and to expand $(3x - 2y + 4z)^2$. <i>[Ch 4: Exploring Algebraic Identities]</i>	5
	OR The diameter of a circle is 26 cm. A chord of length 24 cm is drawn in the circle. Find the distance of the chord from the centre. Also, a cyclic quadrilateral ABCD is inscribed in this circle with $\angle A = 75^\circ$ and $\angle B = 110^\circ$. Find the measures of $\angle C$ and $\angle D$. <i>[Ch 5: I'm Up and Down, and Round and Round]</i>	
33	A city has two main roads which cross each other at the centre of the city. These two roads are along the North-South (N-S) direction and East-West (E-W) direction. All the other streets of the city run parallel to these roads and are 200 m apart. There are 10 streets in each direction. (i) Using 1 cm = 200 m, draw a model of the city in your notebook. Represent the roads/streets by single lines. (ii) There are street intersections in the model. Each street intersection is formed by two streets—one running in the N-S direction and another in the E-W direction. Each street intersection is referred to in the following manner: If the second street running in the N-S direction and 5th street in the E-W direction meet at some crossing, then we call this street intersection (2, 5). Using this convention, find: (a) how many street intersections can be referred to as (4, 3). (b) how many street intersections can be referred to as (3, 4). <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	5
34	A student has ₹500 in her savings bank account. She gets ₹150 every month as pocket money. Construct a table of values showing the total amount she will have at the end of the first 4 months. Find a linear expression to represent the amount she will have in the n^{th} month. Explain why this represents linear growth. <i>[Ch 2: Introduction to Linear Polynomials]</i>	5
Section E Attempt 3 out of 4.		



Q.NO.	QUESTIONS	MARKS
35	Case Study: Room Layout with Coordinates Reiaan is planning his room layout on a coordinate plane. The room door extends from $D_1 (9.5, 0)$ to $R_1 (11.5, 0)$. The bathroom door extends from $B_1 (0, 1.5)$ to $B_2 (0, 4)$. A rectangular study table has three of its feet at the points $(8, 9)$, $(11, 9)$ and $(11, 7)$. Reiaan wants to verify the dimensions of each item and check whether the bathroom door is wider or narrower than the room door. He also needs to find the position of the fourth foot of the table to complete the layout. (i) What is the width of the room door? (ii) What are the coordinates of the fourth foot of the rectangular study table? (iii) Find the width of the bathroom door. Is the bathroom door narrower or wider than the room door? (iv) Both B_1 and B_2 have x-coordinate equal to 0. What does this indicate about the position of the bathroom door with respect to the y-axis? <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	5
	OR Case Study: City Map Coordinates A student is creating a 2-D map of a city center, inspired by how early Arabic maps used the Ujjayinī meridian as a zero-longitude reference. On the map, the main intersection is chosen as the point of intersection of the axes, called the origin. The horizontal road is the x-axis and the vertical road is the y-axis. A cafe is located 3 units to the right and 2 units upwards from the origin. A library is located 4 units to the left and 5 units upwards from the origin. A park is located 2 units to the left and 3 units downwards from the origin. The student marks these locations using coordinates. (i) What are the coordinates of the origin? (ii) The park is located 2 units to the left and 3 units downwards from the origin. In which quadrant does the park lie? (iii) The cafe is at $(3, 2)$. A shop is located at $(7, 2)$. Find the distance between the cafe and the shop. (iv) A bookstore is located on the y-axis, 6 units upwards from the origin, and a florist is located on the x-axis, 6 units to the right of the origin. Are the coordinates of the bookstore and the florist the same? Explain. <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	



Q.NO.	QUESTIONS	MARKS
36	Case Study: Simplifying Rational Expressions In an algebra workshop, students learn to simplify rational expressions by factorising numerators and denominators. The teacher presents the expression $(x^2 - 7x + 12)/(5x^2 + 5x - 100)$. To factorise $x^2 - 7x + 12$, students find two numbers with sum -7 and product 12, yielding $(x - 3)(x - 4)$. For the denominator, they first factor out 5 to get $5(x^2 + x - 20)$, then find two numbers with sum 1 and product -20, yielding $5(x - 4)(x + 5)$. Common factors are then cancelled since the denominator is not zero. (i) What two numbers have a sum of -7 and a product of 12? (ii) Factorise $5x^2 + 5x - 100$ completely. (iii) Simplify the rational expression $(x^2 - 7x + 12)/(5x^2 + 5x - 100)$, assuming the denominator is not zero. (iv) Why can the factor $(x - 4)$ be cancelled during simplification of this rational expression? <i>[Ch 4: Exploring Algebraic Identities]</i>	5
37	Case Study: Monthly Billing and Linear Relationships A community centre charges a fixed annual fee and an additional cost per session for using the tennis court. A member observed that when she used the court for 8 sessions, her total bill was ₹600. When she used it for 12 sessions, her total bill was ₹840. The total bill y depends on the number of sessions x according to the linear relation $y = ax + b$. In this equation, a represents the slope of the line and b represents the y-intercept. The member wants to determine the values of a and b to understand the billing structure. (i) What does the value of 'a' represent in the equation $y = ax + b$? (ii) Find the values of a and b. (iii) At what point does the graph of this equation cut the y-axis? (iv) What does the value of b represent in the context of the community centre? <i>[Ch 2: Introduction to Linear Polynomials]</i>	5

